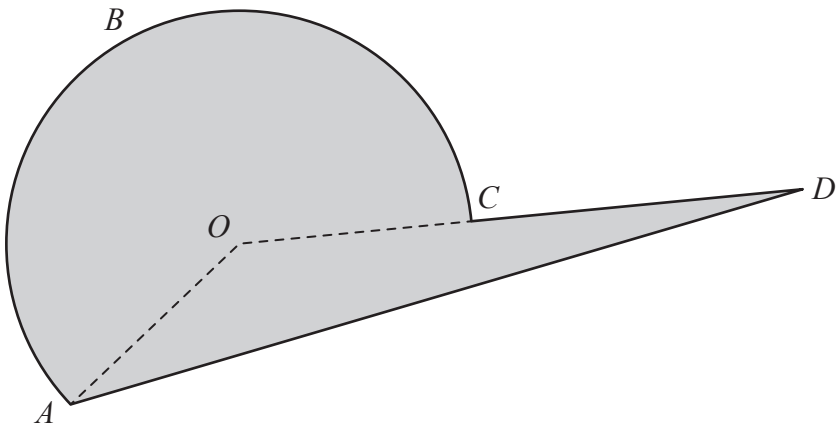
16 Here is a shaded shape $ABCD$.



The shape is made from a triangle and a sector of a circle, centre O and radius 6 cm.
OCD is a straight line.

$AD = 14$ cm
Angle $AOD = 140°$
Angle $OAD = 24°$

Calculate the perimeter of the shape.
Give your answer correct to 3 significant figures.

Major angle = 360° − 140° = 220°

Whole circumference = 2π × 6 = 12π cm

	Degrees	Arc (cm)	
	360	12π	
×11/18			×11/18
	220	$\frac{22\pi}{3}$	

1 mark
Scale the full circle to the major arc.

Sine rule: $\frac{OD}{\sin 24°} = \frac{14}{\sin 140°}$

1 mark
Correct sine-rule relationship.

$OD = 14 \sin 24° \div \sin 140° = 8.858778416...$

1 mark
Find OD.

Perimeter = 23.03834613... + 14 + 8.858778416... − 6

1 mark
Add the major arc, AD and CD = OD − 6.

1 mark
Correct perimeter (cm).

39.9 ..... cm

(Total for Question 16 is 5 marks)

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